

Kumar Babu Boddu

Mechanical Design Engineer
CAD & Product Design Support

✉ kumarbabu20211j@gmail.com

🔗 linkedin.com/in/kumar-babu-79a443306

CAD Modeling | Reverse Engineering | Manufacturing Drawings | Assembly Design

Tools

Creo | SolidWorks | OnShape

Expertise

GD&T | DFM/DFA | Tolerance Stack-up | Mechanism Design

Available for remote mechanical design and product development projects

Deliverables: 3D CAD Models | 2D Manufacturing Drawings | BOM | STEP/IGES Files

Type C Female Connector

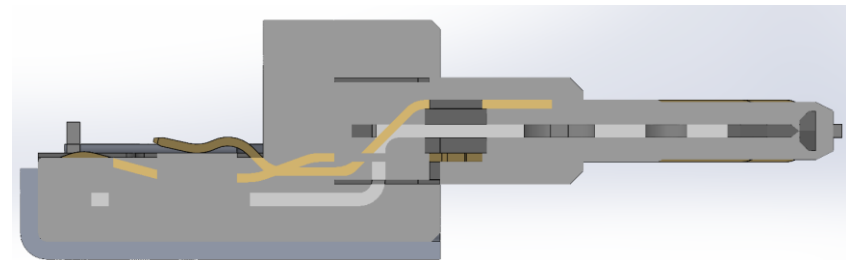
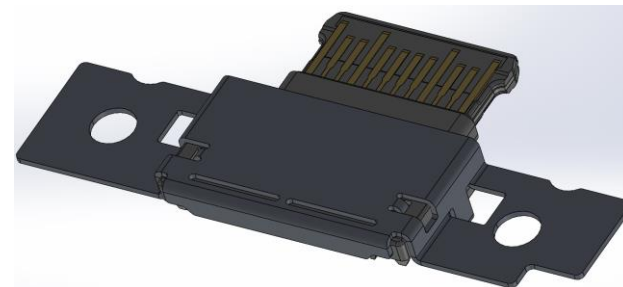
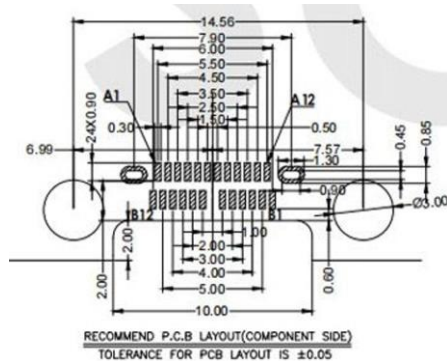
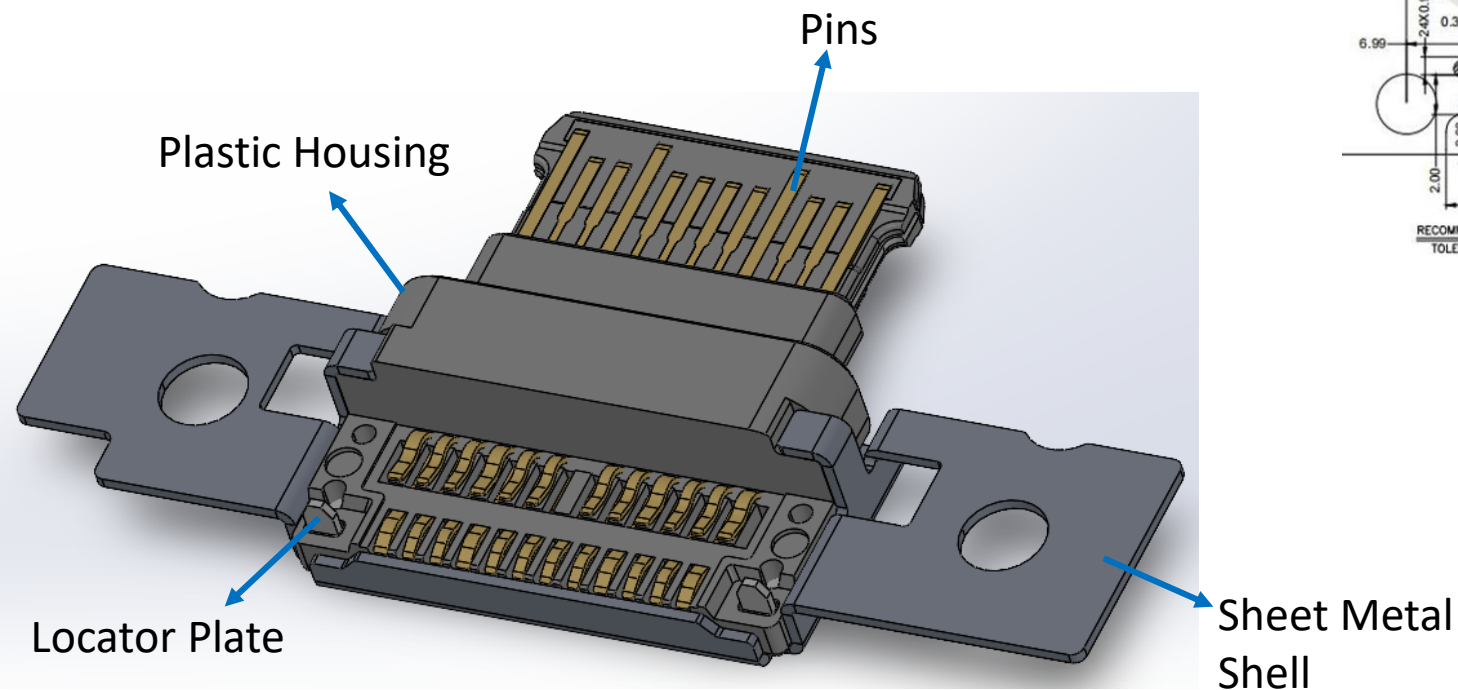
(Electromechanical Design Project)

Description:

Client required a compact 0.5 mm pitch connector model from limited 2D reference for integration into PCB assembly, designed a compact high-density female connector with integrated alignment and retention features, optimized for reliable mating and manufacturability in constrained PCB environments.

Objectives:

- Reverse-engineered connector geometry from reference hardware and drawings.
 - Built fully parametric 3D CAD model with manufacturable tolerances.
 - Developed assembly structure and production-ready drawings.
- 



Internal contact path and retention features

Key challenges: maintaining pin alignment at 0.5 mm pitch and ensuring repeatable mating under tight tolerance stack-up.

Compact high-density board-to-board connector designed for reliable electrical contact and repeatable mating in limited PCB space.

Constraints:

≤ 9 mm height

0.5 mm pin pitch

Alignment tolerance: ±0.05 mm (PCB layout constraint)

0.25 mm SUS304 sheet metal, designed for stamping & bending

Contact resistance ≤ 30 mΩ

Reverse Engineering Strategy:

Reverse-engineered connector geometry from 2D reference drawings. Critical dimensions such as pin pitch (0.5 mm), overall envelope, and mounting features were extracted directly, while internal geometries were inferred using symmetry and standard connector design practices.

Assembly Architecture :

Plastic Housing

- Alignment + insulation
- Snap-fit / retention features

Pins (Contacts)

- Electrical conduction
- Spring geometry for contact force

Sheet Metal Shell

- Mechanical support
- PCB mounting + shielding

Locator Plate

- Ensures positional accuracy

Key Design Decisions:

Pin Pitch (0.5 mm)

Selected 0.5 mm pitch to achieve high contact density while maintaining manufacturability.

Required precise alignment features due to tight spacing.

Alignment Strategy

Integrated locator features and tight PCB tolerance (± 0.05 mm) to ensure reliable mating and prevent pin damage.

Sheet Metal Thickness (0.25 mm)

Chosen to balance structural rigidity with manufacturability in stamping and bending operations.

Failure Modes & Mitigation:

Potential Failure Modes:

Pin misalignment → bending or damage during mating

Wear over cycles → increased contact resistance

Over-insertion / misalignment → mechanical failure of housing

Mitigation Strategies:

Integrated **locator features** for precise alignment

Tight tolerance control (± 0.05 mm)

Optimized **pin geometry** for flexibility and durability

Material selection for wear resistance

Failure modes were considered during design to ensure reliability in high-density, precision mating conditions.

CAD Modeling Strategy:

Developed a parametric CAD model enabling easy modification of pin count and pitch using linear pattern features and constrained assemblies.

- Parametric modeling
- Pattern features (for pins)
- Assembly constraints

Validation / Fit Check:

Section view

Interference check

Alignment verification

Manufacturing Considerations:

- Designed housing with uniform wall thickness and draft angles for injection molding and avoid defects.
- Sheet metal components considering bending limits and forming constraints to ensure manufacturability in stamping processes.

Plastic Housing

Injection molding (draft angles, wall thickness)

Pins

Stamping + forming

Shell

Sheet metal bending

Design Approach:

Followed an iterative design approach with validation at each stage to ensure alignment with requirements and assembly compatibility.

Deliverables:

- Accurate 3D model
- Assembly + drawings
- Manufacturable design

Learnings:

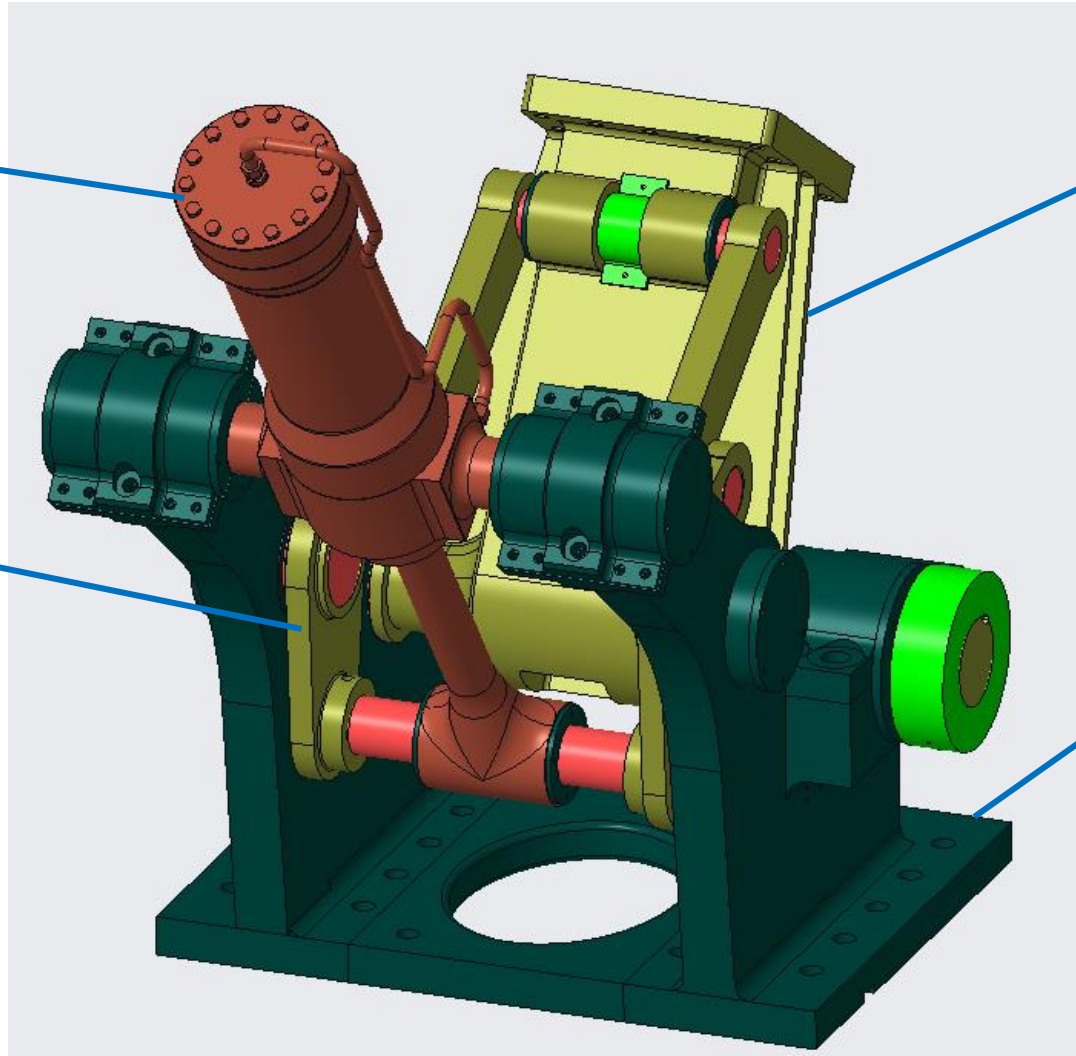
- Handling tight tolerances (± 0.05 mm)
- Challenges of high-density connectors
- Importance of alignment in mating systems

Note: The designs presented are original or recreated representations. No confidential data from previous employers or clients is disclosed.

Electro-Hydraulic Fixture Assembly

(EHS Shaft System)

Description: Electro-mechanical fixture with shaft-driven linkages and hydraulic interfaces, designed for controlled motion and manufacturability.



Piston Eye
Shaft Assembly

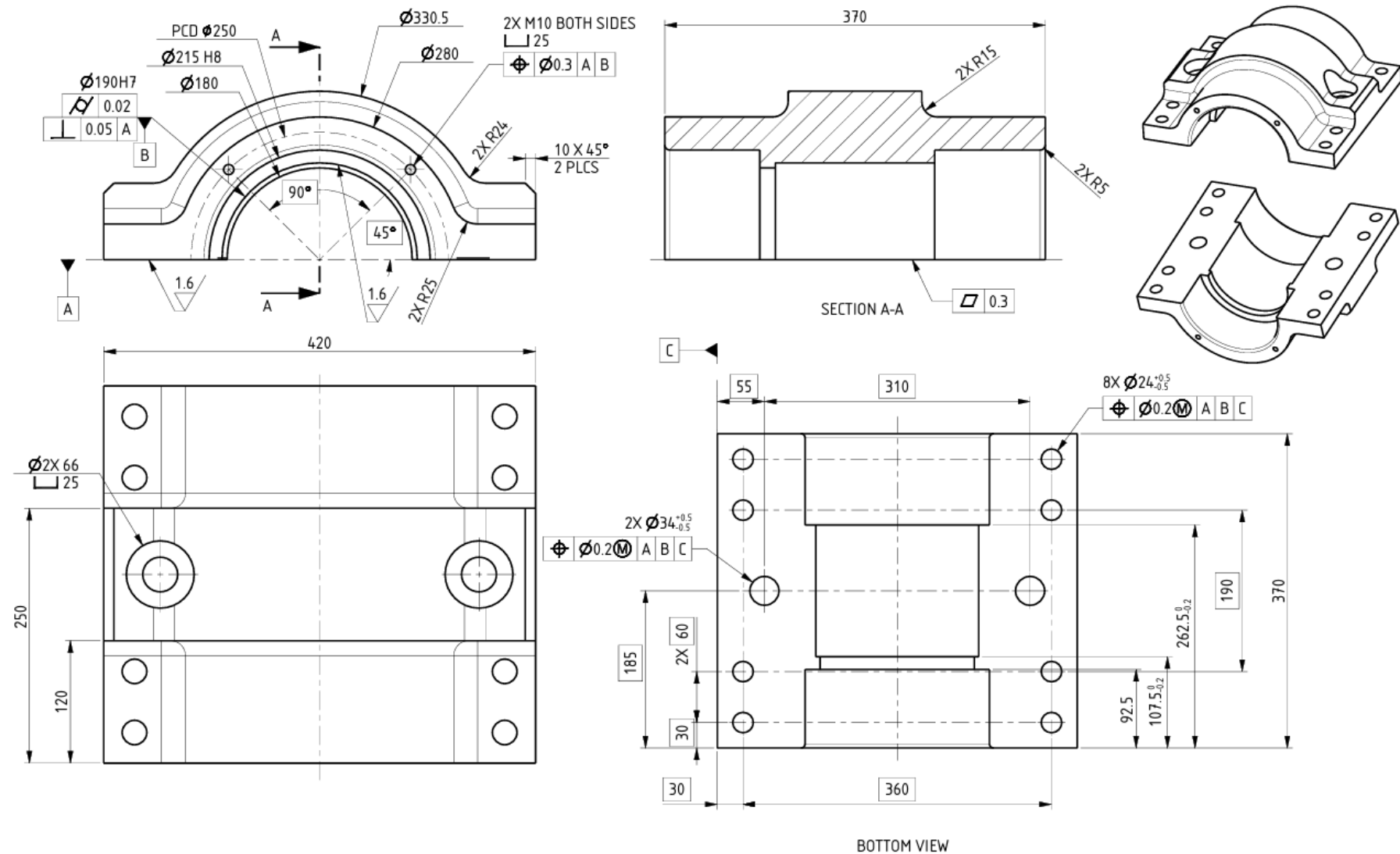
Crank Shaft
Link Assembly

EHS Shaft
Assembly

Base Plate
Assembly



Exploded Assembly View



Key GD&T Features:

- Position tolerance (Ø0.2) ensures precise alignment of mounting holes for proper assembly fit Datum structure (A, B, C) defines assembly reference system.
- Surface finish (Ra 6.3) ensures proper contact surfaces.
- H7/p6 fit ensures secure press-fit for bearing support.

These tolerances ensure reliable assembly, reduce misalignment, and improve overall system performance.

NOTES:
 1. UNSPECIFIED TOLERANCE SHALL BE ± 0.1 MM.
 2. UNSPECIFIED SURFACE FINISH SHALL BE 6.3 MICRONS.
 3. MATERIAL: SUPER DUPLEX STAINLESS STEEL: S32550 (ASTM A240M).
 4. Ø190 BUSH OD TO BEARING SUPPORT BORE - PRESS FIT (H7/p6).
 PRESS USING HYDRAULIC PRESS ONLY. NO HEATING UNLESS APPROVED.

SP-01 BEARING SUPPORT-01: RD-DG-3-50-20-RM-32

Manufacturing drawing with GD&T applied to control critical interfaces, ensuring precise alignment and reliable assembly performance.

Project Scope:

Design modification and validation of electro-hydraulic fixture assembly focusing on mechanism performance, manufacturability, and assembly accuracy.

◆ Contribution:

- Redesigned and modified 3D assemblies in Creo using imported STEP data.
- Corrected assembly interfaces ensuring proper fit, clearance, and motion.
- Developed detailed manufacturing drawings with GD&T.
- Generated BOM structures for assemblies and sub-assemblies.
- Designed and validated mechanisms including shafts, pivots, and linkages.

◆ Engineering Highlights:

- Applied GD&T for critical features (shaft bores, bush interfaces).
- Performed tolerance stack-up analysis to ensure functional assembly.
- Implemented DFM/DFA principles to improve manufacturability and assembly.
- Validated kinematic motion for functional reliability.

🔄 Design Approach:

Followed an iterative design approach with validation at each stage to ensure alignment with requirements and assembly compatibility.

🔥 Impact:

- Improved assembly fit, reducing rework during manufacturing.
- Enhanced functional reliability and motion consistency of mechanism.

Role: Engineering Consultant - Mechanical Design

Tools: Creo, SolidWorks

Standards: GD&T (ISO)

Domain: Mechanical Fixture Design

Before vs After:

Before: Misalignment issues

After: Corrected interfaces + improved fit

📁 Deliverables Provided:

3D CAD (STEP, native format)

2D drawings with GD&T

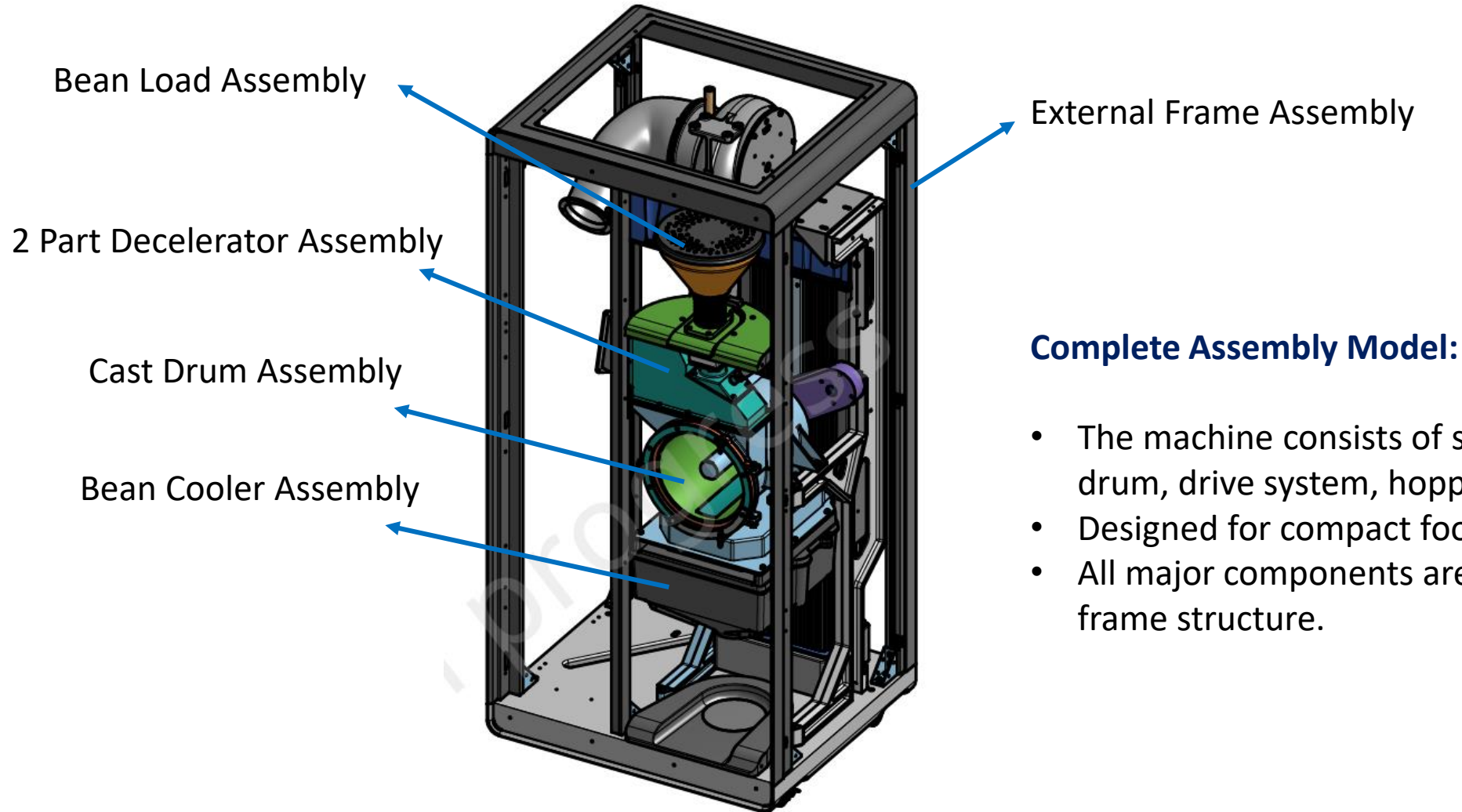
BOM structure

Assembly instructions / exploded view

Coffee Roasting Machine

Mechanical Design / Thermal & Structural Considerations

- **Designed** using OnShape
- **Focus:** Assembly design, sub-assemblies, and manufacturability
- **Scope:** Concept to detailed 3D assembly



Complete Assembly Model:

- The machine consists of structural frame, roasting drum, drive system, hopper, and cooling section.
- Designed for compact footprint and accessibility.
- All major components are integrated within a rigid frame structure.

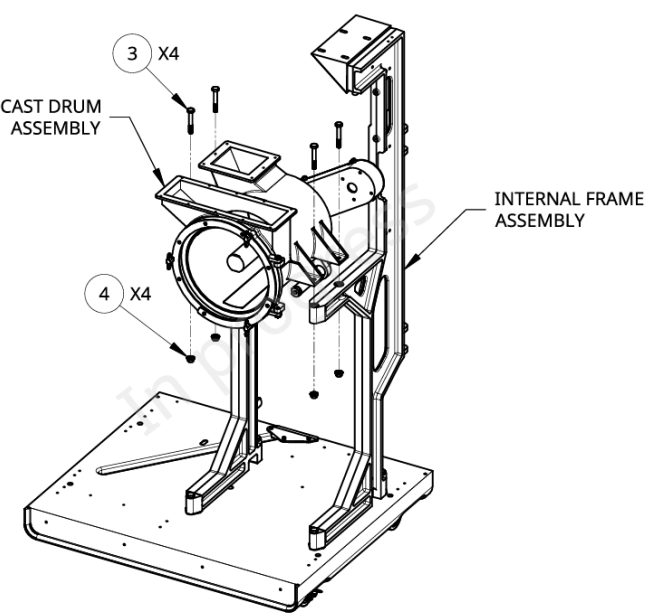
Project Overview: This project involves the design of a compact coffee roasting machine with emphasis on mechanical architecture, component integration, and manufacturability.

Objectives:

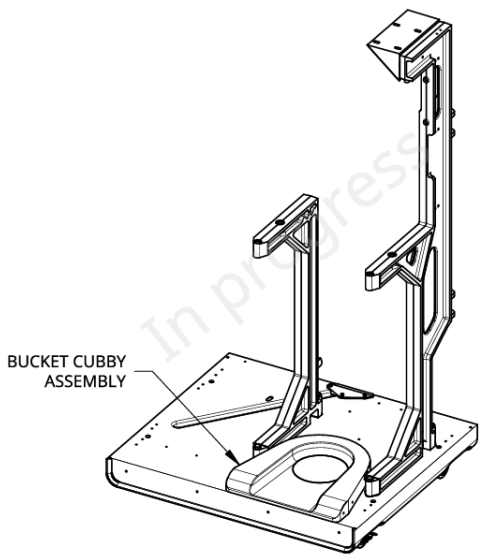
- Develop a complete mechanical assembly.
- Design functional sub-systems (drum, drive, frame, hopper).
- Ensure ease of assembly and maintenance.

The system is divided into the following sub-assemblies:

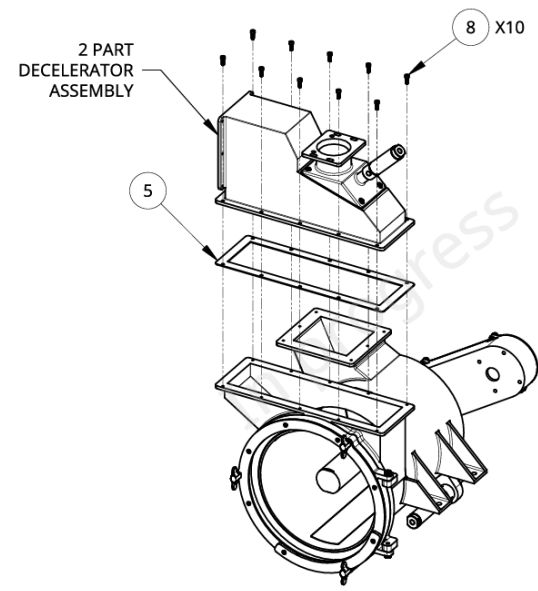
CAST DRUM ASSEMBLY



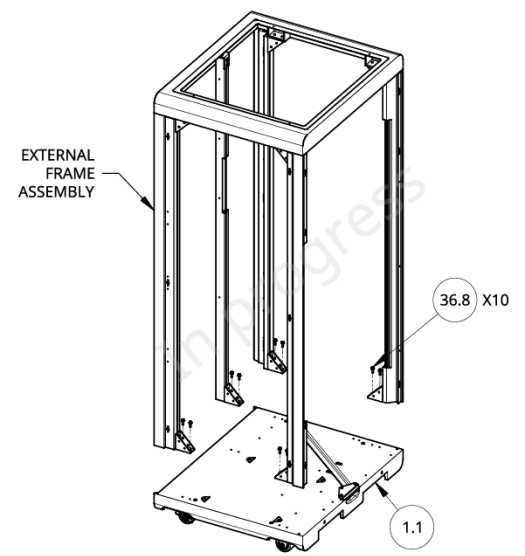
BUCKET CUBBY ASSEMBLY



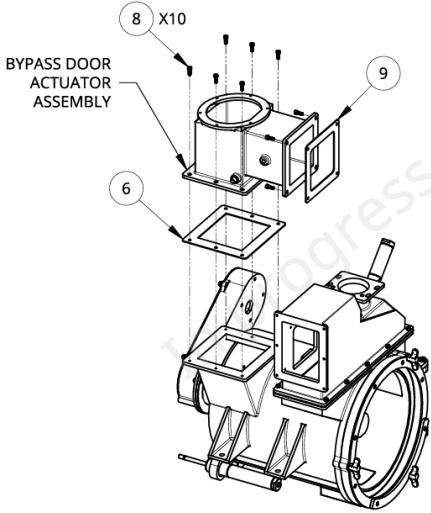
2 PART DECELERATOR ASSEMBLY



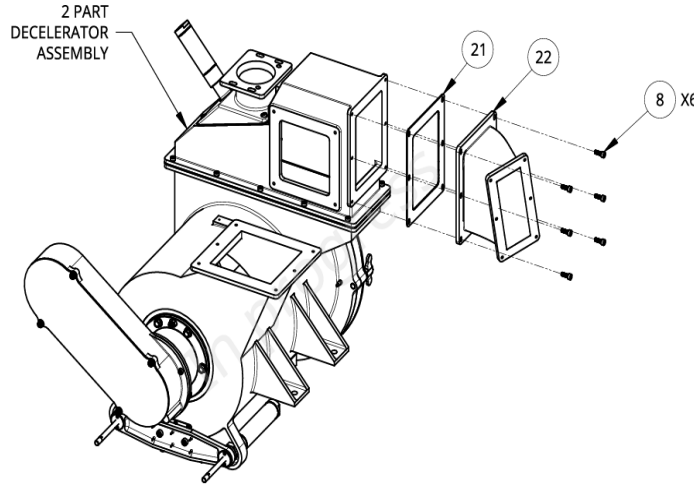
EXTERNAL FRAME ASSEMBLY



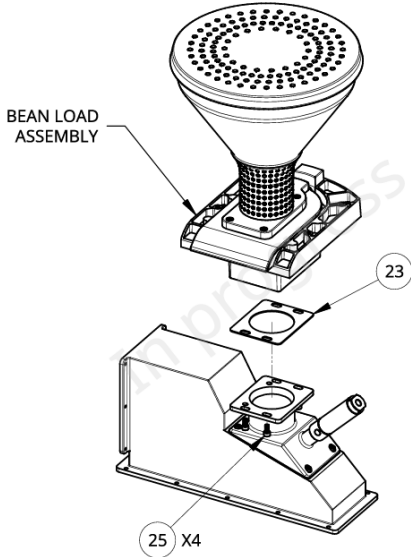
BYPASS DOOR ACTUATOR ASSEMBLY



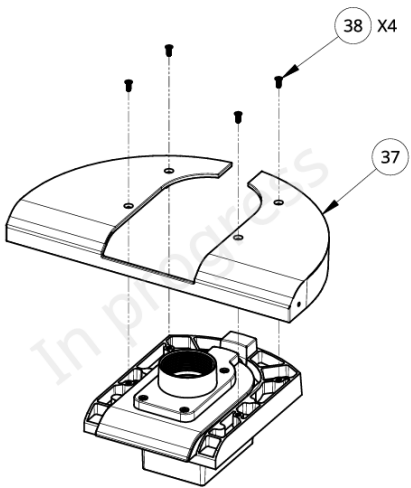
TRANSITION DUCT



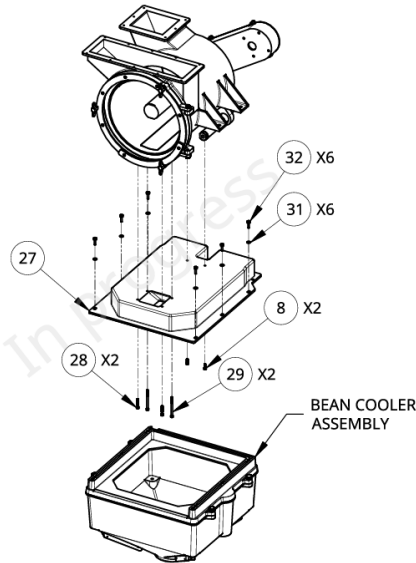
BEAN LOAD ASSEMBLY



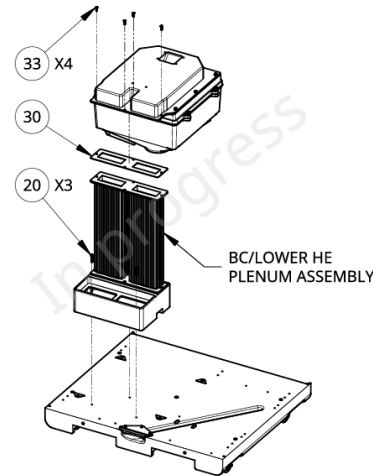
HOPPER CUBBY BASE



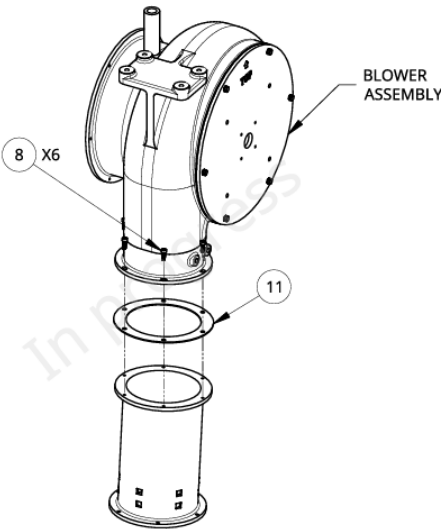
BEAN COOLER ASSEMBLY



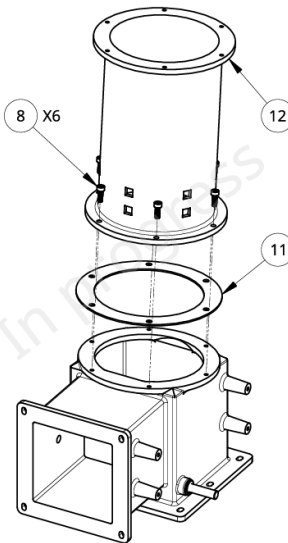
BC/LOWER HE PLENUM ASSEMBLY



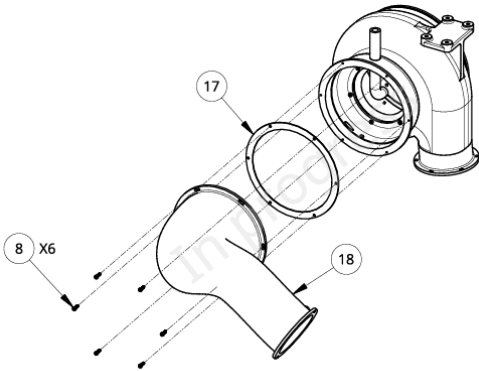
BLOWER ASSEMBLY



AUX 4KW HEATER DUCT



ONE PIECE SCRUBBER DUCT



Exploded View Strategy:

- Individual sub-assemblies are exploded for clarity
- Part relationships
- Assembly sequence
- Maintenance access

Standard Components

- Bearings
- Fasteners
- Motor

Design for Manufacturing (DFM)

- Frame: Welded or bolted structure
- Drum: Sheet metal fabrication / machining
- Shaft: Machined component
- Hopper: Sheet metal forming

Design for Assembly (DFA)

- Modular sub-assemblies
- Easy access to critical components
- Reduced assembly complexity

Design Approach

Followed an iterative design approach with validation at each stage to ensure alignment with requirements and assembly compatibility.

Manufacturing Considerations:

- Frame → welded/bolted
- Drum → fabrication
- Shaft → machining

Deliverables Provided:

3D CAD (STEP, native format)

2D drawings with GD&T

BOM structure

Assembly instructions / exploded view

Key Learnings:

- Improved understanding of assembly design
- Exposure to multi-component integration
- Practical knowledge of mechanical systems
- Importance of manufacturability in design

Outdoor Chair

Reverse Engineering / CAD Modeling & Assembly

Overview

Reverse-engineered an outdoor chair from reference images and developed a complete 3D CAD model with assembly structure and manufacturing drawings.

Key Work

- Recreated tubular frame geometry from visual references
- Modeled woven seat/back structure (HDPE/Nylon assumption)
- Built full assembly with frame, mesh, and armrests
- Generated manufacturing drawings and exploded view

Manufacturing Insight

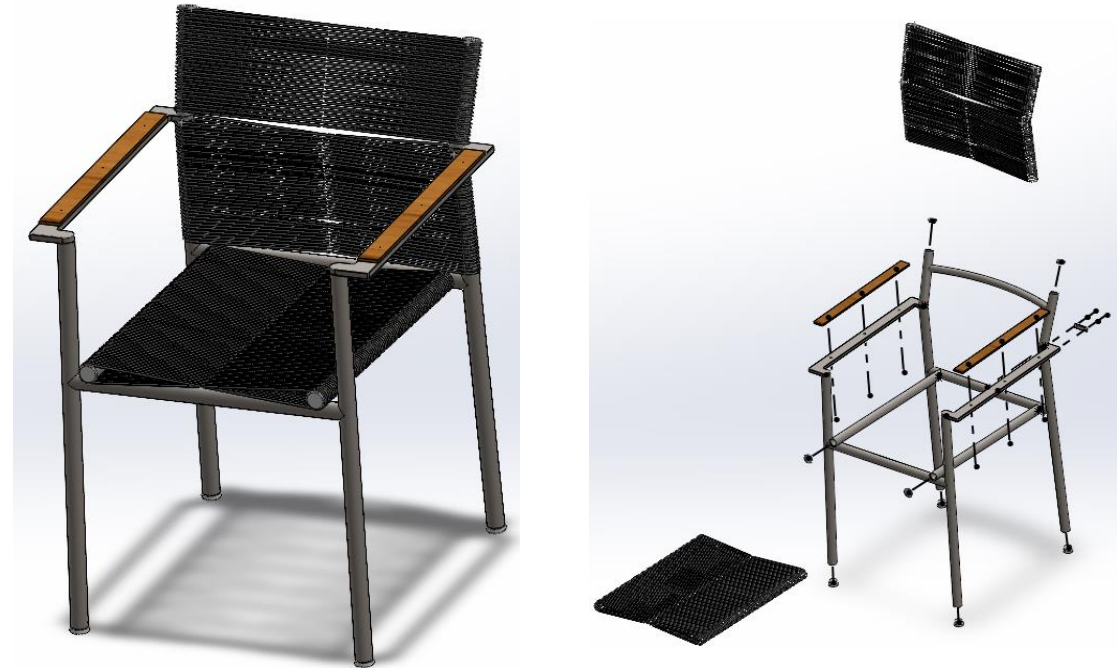
- Frame: tube bending + welding
- Seat/Back: woven polymer mesh
- Armrest: wood machining + fastening

Design Approach

Used an iterative modeling approach to refine proportions, ensure structural stability, and maintain realistic assembly relationships. Designed frame geometry to maintain stability and load distribution under typical seating conditions.

Deliverables

- 3D CAD model (SolidWorks)
- 2D drawings
- Exploded assembly view



Professional Experience Highlights

Safety Switch Components

- Designed and modeled components for safety switch assemblies
- Created manufacturing drawings with GD&T
- Ensured compliance with functional and safety requirements

Home Appliances

- Contributed to design of mechanical components for consumer appliances
- Focused on manufacturability and assembly considerations
- Developed 3D CAD models and detailed drawings

RFC Machines (Industrial Equipment)

- Worked on mechanical assemblies for industrial machinery
- Supported design modifications and production drawings
- Ensured proper fit, alignment, and functionality

Note: Detailed visuals are not included due to confidentiality constraints.

Contact

Kumar Babu Boddu

Mechanical Design Engineer

CAD & Product Design Support

✉ kumarbabu20211j@gmail.com

☎ Phone / WhatsApp: +91-9618497524

🔗 [linkedin.com/in/kumar-babu-79a443306](https://www.linkedin.com/in/kumar-babu-79a443306)

🌐 Based in: Hyderabad, India

Available for remote mechanical design and product development projects

Open to both short-term and long-term collaborations